

DATA STRUCTURE

Binary Search Tree

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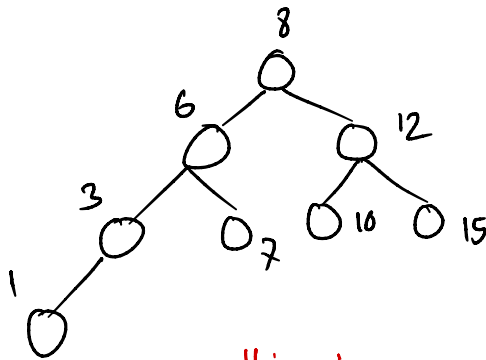
Binary Search Tree (BST)

- must be a binary tree
- $\text{left} < \text{root} < \text{right} \rightarrow$ must follow this

* equal elements must go to either left or right.

if in left $\rightarrow$ left biased

if in Right $\rightarrow$ Right biased

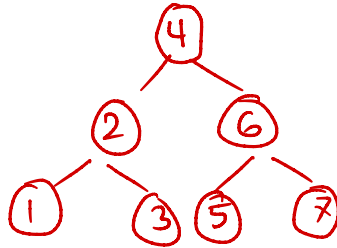


this is an example of BST

▣ Insertion Time $\rightarrow$ if the value is bigger than the root traverse right and recurse

otherwise traverse left and recurse...

* we will get a sorted stream by traversing Inorder



inorder $\rightarrow$ 1 2 3 4 5 6 7 $\rightarrow$ Sorted

* height of BST $\rightarrow \log(n)$

Deletion of BST

- is the node leaf node?

if yes $\rightarrow$ Delete that node. No issue.

if no $\rightarrow$ Ask next ques ...

- does it has 1 child ?

- if yes $\rightarrow$ join the child subtree to
it's grand parent...

- if no $\rightarrow$ there are two steps ...

1. swap value of that node
with the right most leaf node
(biggest in that subtree).

2. now delete the leaf node .

- no issue.

that's how we can delete a node in BST

